

WebExpert: Expert-Level Web Search Agent for Professional Domains

Critic-Guided Expert Experience · Schema-Light Facet Induction · Preference-Optimized Planning

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EM BOOST

+1.5~3.6

pp over strongest browsing baseline

QP@3 ↑

61.8

vs. 49.3 (WebThinker baseline)

PAGE HOPS ↓

5.2

vs. 8.1 — faster & cheaper

1 MOTIVATION & CONTRIBUTIONS

Problem. Generic web-search agents issue **vague, keyword-style queries** (e.g. “new-energy subsidy policy impact on bus companies”) → **token waste & insufficient evidence**.

Key insight. Domain experts rewrite the same question with **exact facets**: **Zhejiang Province · 2023–2025 · New-Energy Cars · Local Subsidies · Purchase-Tax Exemptions · Document No.** — hitting **authoritative, grounded** evidence instantly.

Our contributions.

- ① **Critic-guided expert-experience** chain that distills QA logs into reusable, rule-level knowledge.
- ② **Schema-light facet induction** — no handcrafted ontology.
- ③ **Preference-optimised planning** with dual loss $\mathcal{L}_{\text{plan}} + \alpha \cdot \mathcal{L}_{\text{ret}}$.
- ④ **SOTA on 4 benchmarks** (GAIA, GPQA, HLE, WWQ) with **fewer page hops**.

2 WEBEXPERT FRAMEWORK (FIG. 1)

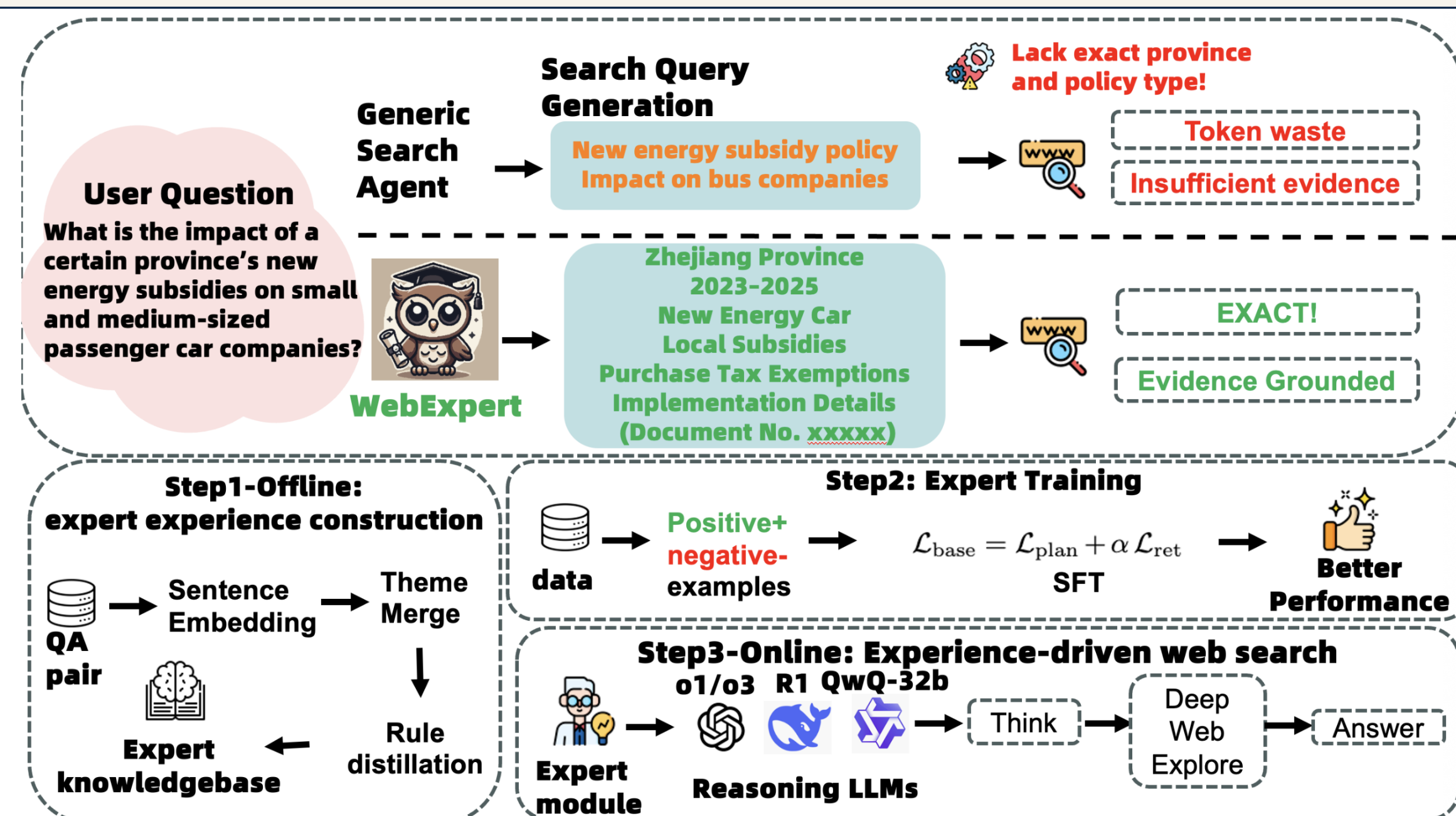


Fig. 1 — Generic agent vs. WebExpert · 3-step pipeline (paper).

Top — Side-by-side comparison.

A generic agent sends **vague keyword queries** (e.g. *new-energy subsidy · bus companies*) → **Token waste · Insufficient evidence**. WebExpert rewrites the same question with **exact facets** from the expert memory (Zhejiang Province 2023–2025 · New-Energy Cars · Local Subsidies · Purchase-Tax Exemptions · Document No. xxxx) → **EXACT! Evidence grounded**.

Bottom — Three-step pipeline.

Step 1 · Offline — Expert-experience construction. QA → sentence embedding → theme merge → rule distillation → expert KB.

Step 2 · Training — Expert training. Curate positive / negative pairs from the KB and fine-tune the reasoning LLM via $\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{plan}} + \alpha \cdot \mathcal{L}_{\text{ret}}$ (SFT + preference loss on retrieval).

Step 3 · Online — Experience-driven web search. Given the user query, the **Expert module** retrieves top-k rules, rewrites the query, then the reasoning LLM (o1 · o3 · R1 · QwQ-32B) Think → Deep-Explore → Answer.

3 CRITIC-GUIDED EXPERIENCE EXTRACTION

① Question Harvesting

Collect (q, a, R, C) tuples; normalize via paraphrase mining & schema-free **delexicalization** → canonical intents q■.

② QA Multi-view Clustering

Jointly encode questions / answers / (q,a) pairs; **HDBSCAN + UMAP + BERTopic** with soft assignments to allow overlapping intents.

③ Evidence Aggregation

BM25 + dense retrieval on answers & rationales; **MMR reranking** for source-level diversity and quote-level de-dup.

④ Contradiction-Aware Summary

DeepSeek-R1 summarizer emits a rule with *conditions · core guidance · edge cases · failure modes*; entailment filter drops self-contradictory claims.

⑤ Facetization & Normalization

Facetize each rule into (**time, region, policy, L2-industry**); corpus-stat filter (e.g. “CFA”, “FDA”) + shallow taggers + LLM disambiguation; attach coverage / confidence / provenance.

⑥ Continuous Refresh

Versioned experience store with **warm-start clustering** and local merges — stable rule IDs while streaming in new QA evidence.

4 TRAINING OBJECTIVE (QwQ-32B, Full FT)

$$\mathcal{L} = \mathcal{L}_{\text{plan}} + \alpha \cdot \mathcal{L}_{\text{ret}} + \beta \cdot \mathcal{L}_{\text{pref}}$$

α tuned on GAIA val · β swept in {0.1, 0.3, 0.5} · best: $(\alpha, \beta) = (0.5, 0.3)$

DATA

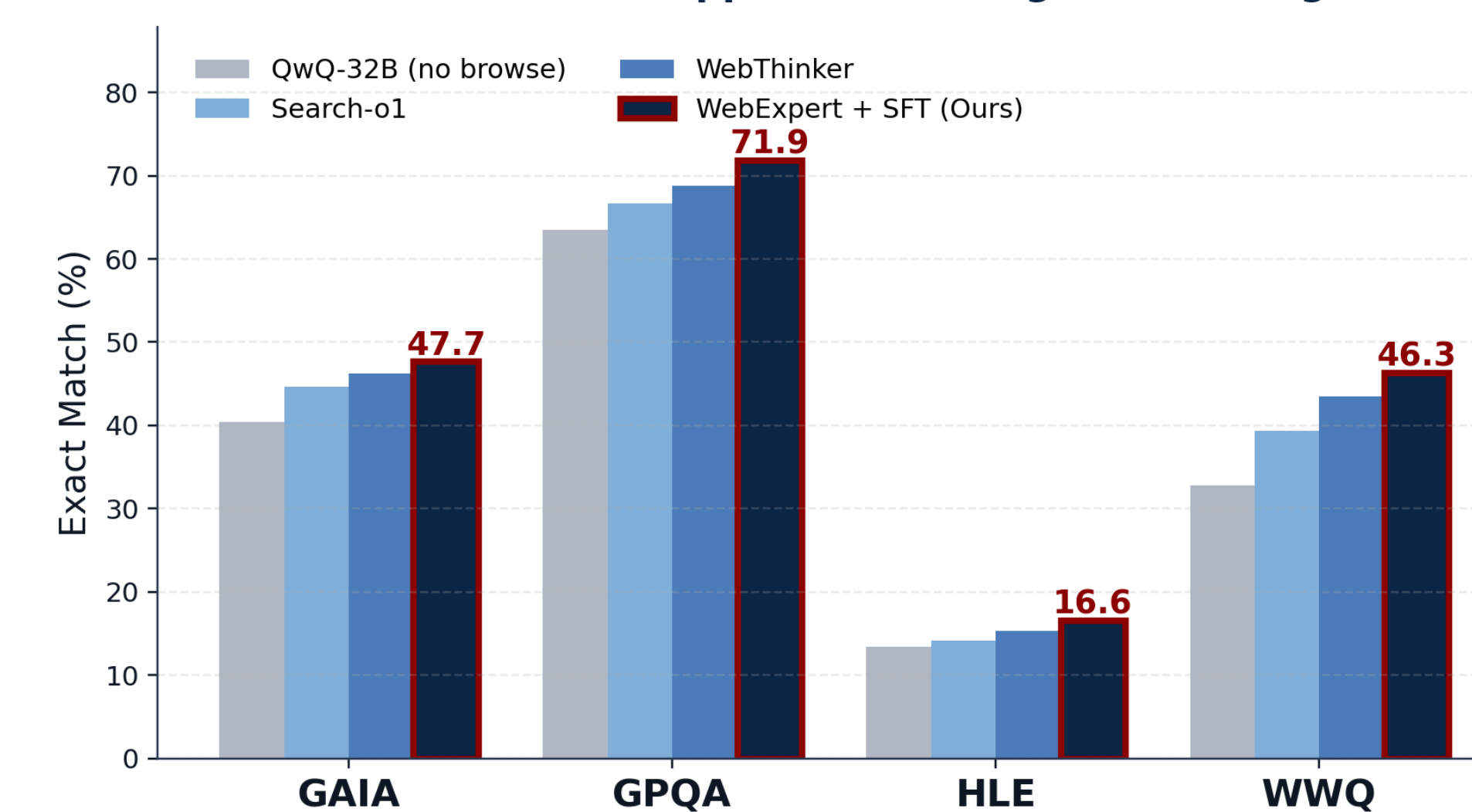
~12k pref pairs · positives = facet-aligned · negatives = off-facet / redundant

TRAINING

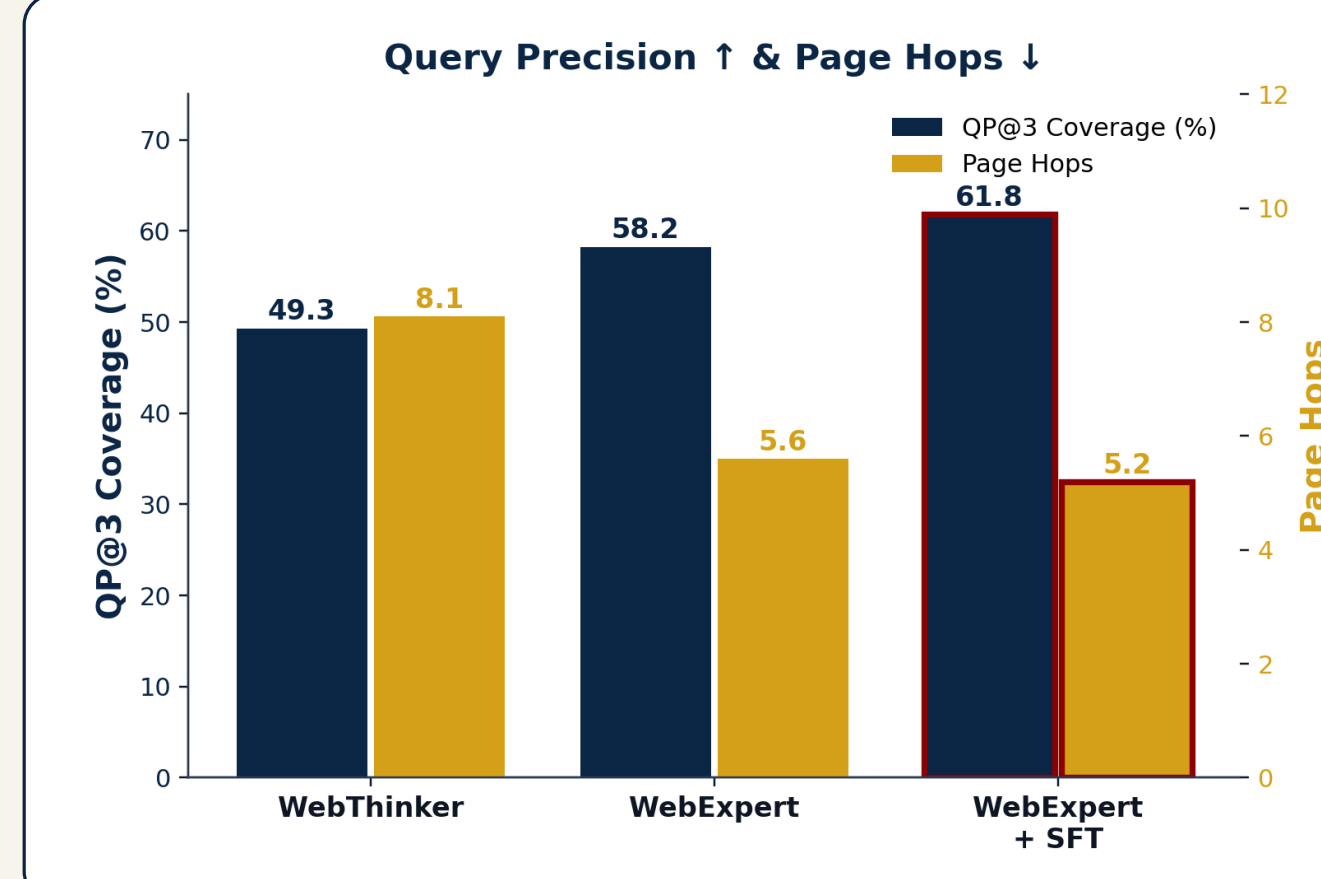
Pai-Megatron-Patch · lr 1e-5 · cosine · beta2 = 0.98 · early-stop on QP@3 + EM

5 MAIN RESULTS — 4 BENCHMARKS

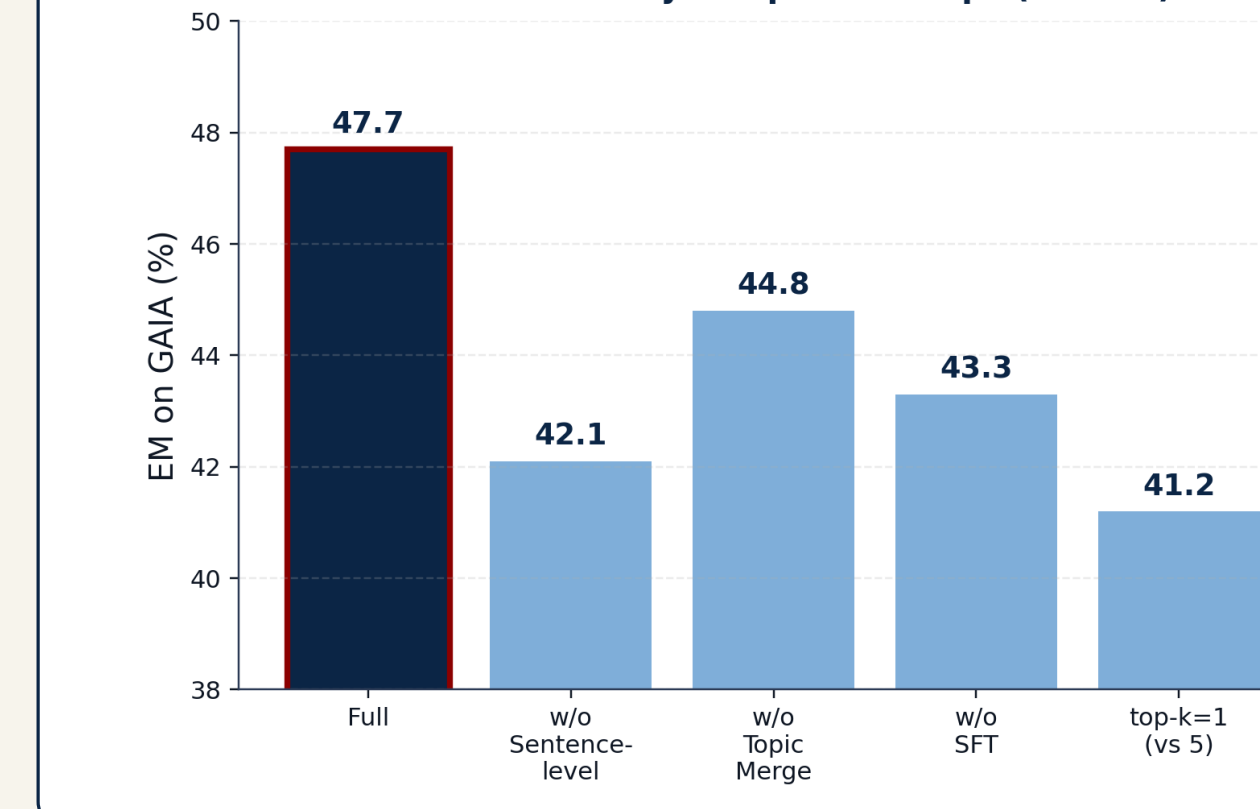
Main Results — +1.5~3.6 pp EM vs. strongest browsing baseline



6 QUERY QUALITY & ABLATION



Ablation — every component helps (Table 3)



KEY TAKEAWAYS

• **Sentence-level retrieval + SFT** contribute the most on GAIA — retrieving **top-5 experiences** balances precision and coverage; **top-1 drops EM by ~6 pp**.

• **Topic merging** is decisive: rule fragmentation cuts EM by ~3 pp and inflates page hops; **MMR + quote-level de-dup** trims redundant evidence.

• **Schema-light facets** (time · region · policy · L2-industry) transfer across finance / biomedicine / pharmacy without hand-written ontologies.

• **nDCG@10 ↑ 4–6 pts** across GAIA / GPQA / HLE / WWQ; Page Hops drop **8.1 → 5.6 → 5.2** (+SFT).

• **Experience gate** ($\theta=0.3$) falls back to generic query-gen when retrieval confidence is low — guards against over-constraint.

• **Leakage stress tests** (entity-random / time-shift / template-remix EM) confirm *gains come from facet grounding, not memorization*.

• Plug-in with **o1 / o3 / R1 / QwQ-32B** reasoning backbones — expert module is frozen at inference.

Expert Experience → Expert Query → Expert Evidence → Expert Answer